

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

I E.Ragul / 192571032 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E. Ragul

Technical Presentation

1. Reactive Agent-Based Traffic Control System

Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis

Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations

Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System

Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

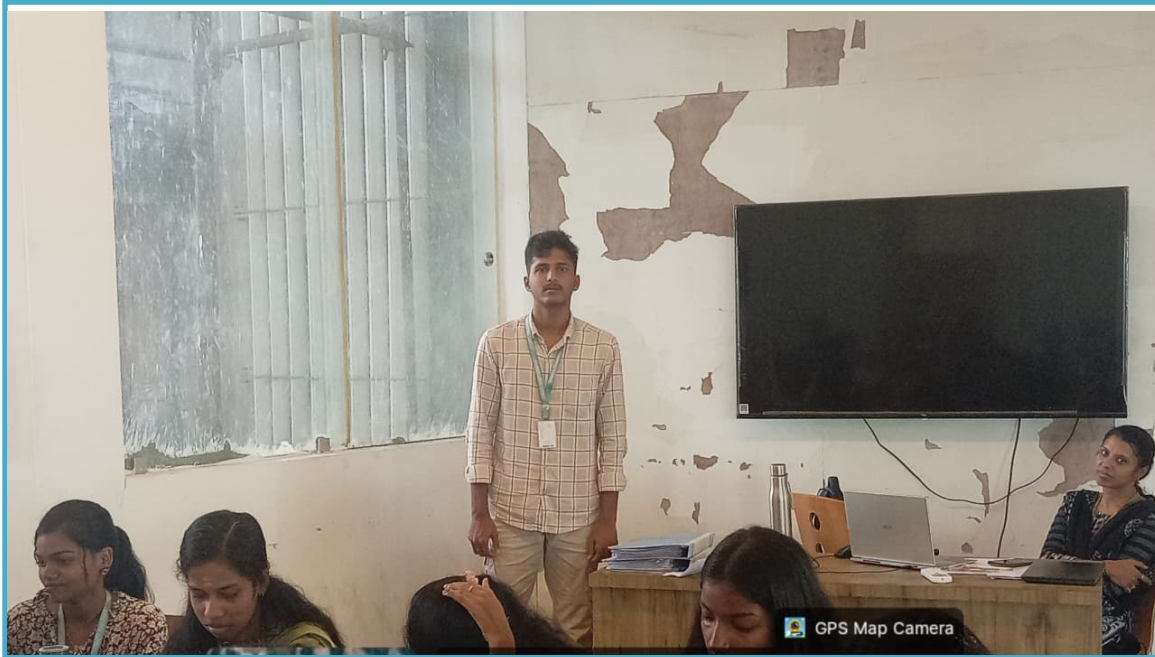
5. Multi-Agent System for Disaster Management

Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Multi-Agent System for Disaster Management



1. Introduction

A Multi-Agent System for Disaster Management is an intelligent system that uses multiple agents to detect, analyze, and respond to disasters. Each agent performs a specific task and communicates with other agents to handle emergency situations.

The main objective is to detect disasters quickly, provide timely warnings, coordinate rescue operations, reduce damage, and save human lives.

2. What is a Multi-Agent System?

A Multi-Agent System is an AI system in which multiple intelligent agents work together to achieve a common goal.

Each agent:

- Observes a specific part of the environment.
- Collects and processes information.
- Makes decisions based on predefined rules.
- Communicates with other agents.
- Takes suitable action.

For example:

IF a sensor agent detects a high water level
THEN send the information to the decision agent.
IF the decision agent confirms a flood
THEN the communication agent sends an emergency alert.

3. System Components

Sensor Agents

- Collect information from sensors and cameras.
- Detect floods, fires, earthquakes, and other disasters.
- Monitor temperature, water level, and environmental conditions.

Decision Agents

- Receive information from sensor agents.
- Analyze the disaster situation.
- Determine the severity of the disaster.
- Decide suitable rescue actions.

Communication Agents

- Send emergency alerts to people.
- Share information with rescue teams.
- Transfer information between different agents.

Rescue Agents

- Locate affected areas.
- Support rescue teams.
- Help identify safe routes and evacuation areas.

Communication Network

- Connects all agents.
- Transfers disaster information in real time.
- Supports coordination between agents.

Simple Flow

Sensors → Sensor Agent → Decision Agent → Communication Agent → Rescue Team → Disaster Response